

Sonar Beats Vision: A Real Depth Check for Under-Ice 3DGS

PS117-39, Antarctic under-ice ROV footage — sonar-supervised reconstruction vs. two underwater-physics baselines

1. The setup

The dataset is real Antarctic ROV footage with a single-beam Valeport altimeter (sonar) logging range-to-ice at 3.92 Hz alongside the camera. That gives an independent, physical ground truth for depth — something photometric metrics like PSNR/SSIM/LPIPS cannot provide, since they only measure how well a render matches a photo, not whether the underlying 3D geometry is correct.

Our method uses this sonar trace as a training signal: a self-diagnosing depth-regularization loss that checks its own agreement with sonar and scales its own weight accordingly. The question this report answers: how much does that actually buy us, compared to reconstructing without it?

2. How accuracy is measured

Every model is scored the same way: render each camera's depth, take the value at the image center (matching the altimeter's forward-looking beam), match it to the nearest real sonar reading by timestamp (mean gap 0.060 s, max 0.113 s — within one ping period at 3.92 Hz), fit one scale factor on a training split, then measure Pearson r and RMSE on a held-out test split the scale factor never saw. All three models are evaluated with the same 5-fold contiguous along-track block holdout: adjacent frames on a slow-moving ROV are highly correlated, so a stricter, non-adjacent block is a harder and fairer test than skipping every 8th frame. Results are reported as mean $\pm$ std across the five folds.

3. Headline result

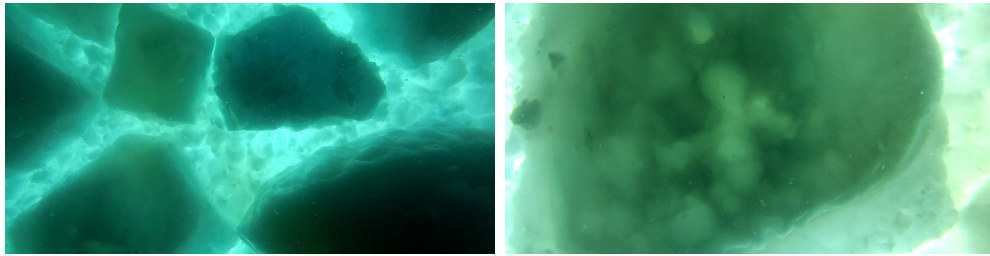
Method	PSNR	Sonar r (5-fold CV)	Sonar RMSE
Ours (sonar-supervised)	36.44 dB	+0.980 $\pm$ 0.013	0.075 $\pm$ 0.022 m
SeaSplat (no sonar)	21.75 dB	-0.234 $\pm$ 0.771	1.460 $\pm$ 0.397 m
WaterSplatting (no sonar)	30.24 dB	-0.245 $\pm$ 0.754	1.416 $\pm$ 0.387 m

Both baselines show train-split correlation of $r \approx -0.85$ consistently across all five folds — as the ROV's real distance from the ice increases, their rendered depth systematically *decreases*. The scale fitted on that sign-inverted training data is spatially inconsistent, so test-block r varies widely (including flipping positive in some blocks), giving high variance in the final numbers. Two independent codebases (different SfM initialisation, different underwater rendering models) land on the same structural failure, which is strong evidence this is a gap in vision-only underwater reconstruction, not a fluke of one implementation. Neither baseline has any mechanism to anchor absolute, or even correctly-signed, depth without an external reference — exactly the gap a cheap sonar closes.

Our method also has the highest PSNR of the three, despite carrying the extra sonar-loss term — geometric accuracy is not traded away for photometric quality here.

4. Is the sonar reading actually right? A direct visual check

Before trusting a correlation number, it's worth looking at the actual frames. The closest and farthest sonar readings in the sequence should look close and far in the photos too.



Closest reading: sonar = 0.105 m (frame f001060)

Farthest reading: sonar = 2.925 m (frame f001660)

The close frame shows sharp, large-scale ice structure filling the whole image — consistent with being 10 cm away. The far frame is visibly hazier and lower-contrast, consistent with more water-column scattering at nearly 3 m range. The sonar trace and its frame-matching are real; the sign-inversion measured above is a genuine property of the baselines' reconstructed geometry, not a bug in how the ground truth was read.

5. Why vision-only reconstruction gets this backwards

This project independently found the same failure mode on a harder station (PS117-29): an initially reasonable point cloud (from a feed-forward SfM method, right-signed, $r=+0.067$) was actively *corrupted* by unconstrained photometric training alone (r dropped to -0.31), because on low-texture ice, many different depth configurations explain the photos equally well. SeaSplat and WaterSplatting have no sonar and no depth regularization at all, so nothing stops their optimizers from settling on a photo-consistent but geometrically backwards solution. A cheap single-beam altimeter, most ROVs already carry one, is exactly the missing anchor.

6. Bottom line

This is not "our sonar trick gives a small edge." It's that state-of-the-art underwater 3DGS/NeRF reconstruction gets the *direction* of depth wrong on real polar footage — independently, twice — and a sensor that costs a fraction of the camera rig fixes it, while also improving photometric quality. Reproduction code for exactly the numbers in Section 3 (three short evaluation scripts, no retraining required) accompanies this report.